Single-image Super-resolution via De-biased Sparse Representation

Jian Pu[1], Yingbin Zheng[2] and Hao Ye[2]
[1] School of Computer Science and Software Engineering, East China Normal University, Shanghai, China
e-mail: jianpu@sei.ecnu.edu.cn
[2] Shanghai Advanced Research Institute, Chinese Academy of Sciences, Shanghai, China
e-mail: zhengyb@sari.ac.cn, yeh@sari.ac.cn

***Abstract*— Sparse representation and dictionary learning of image patches are well-known methods for single-image super-resolution. However, due to the regularization term of sparse-inducing penalties, the solution is usually biased. In this study, we present a de-biasing framework by adding a de-biasing step after sparse representation. Two de-biasing methods with sign consistency and feature consistency are further proposed under this framework. Using a unified proximal gradient method, we can solve the proposed de-biasing methods efficiently. Experiments on real super-resolution datasets validate the effectiveness and robustness of the proposed de-biasing methods.**

***Keywords*— De-biased sparse representation, ℓ_1 regularization, Single-image super-resolution, Dictionary learning**

I. Introduction

Single-image super-resolution is to construct the corresponding high-resolution (HR) image from one given low-resolution (LR) image. The fundamental idea behind is to take advantage of the repetitive structure information (e.g., edges, corners, textures, etc.) of natural images [1]. After cutting each image into a set of small overlapped patches, single-image super-resolution is to discover the relationship between LR and HR image patches. Assuming the LR and HR image patches sharing the same geometry structure, sparse representation is one of the well-known methods to first coding the LR patches and then reconstruct the HR patches [2]. In this super-resolution process, sparse representation is the most critical step. However, for frequently used ℓ_1 regularization, it inevitably brings significant bias of the representation, and thereby diminishes the quality of the final construction of HR image.

There are usually two kinds of methods to eliminate or reduce the impact of the biased estimator in literature. One way is to use relative a small regularization parameter for ℓ_1 regularization. Though intuitive, it is usually tricky to choose such parameter. A very small regularization parameter may lead to dense representation, and thus lose the good properties of sparse representation. The other way is first to identify the support of representation using k nearest neighbors, and then to code the patch using ridge regression [3], neighbor embedding [4], [5]. The main issue of such heuristic methods is the lack of theoretical analysis.

Though ℓ_1 regularization induces bias inevitably, we notice that some good properties (e.g., feature consistency [6] and sign consistency [7], [8]) are ensured under some mild conditions. To retain these properties, we propose a de-biasing framework by adding a de-biasing step after ℓ_1 sparse representation. The de-biasing step is formed as an optimization of the data fidelity with consistency constraints. Furthermore, we present two de-biasing methods with sign consistency and feature consistency under this framework. These proposed methods can be finally optimized using a unified proximal gradient method. In addition, we exhibit the equivalence of sign consistency with generalized Bregman distance minimization and feature consistency with infimal convolution of Bregman distance minimization. Finally, we apply the proposed de-biased methods for single-image super-resolution. Experimental results on two commonly used super-resolution datasets demonstrate the effectiveness and robustness of the proposed de-biasing methods.

The rest of this paper is organized as follows. In Section II, we give a brief review of related super-resolution and dictionary learning methods. Section III presents the detailed de-biasing framework and two methods under this framework. Section IV shows the improved single-image super-resolution method. Section V reports the numerical and visual results on two real image datasets. Finally, Section VI concludes the paper.

II. Related Work

The pioneer patch-based work for single-image super-resolution is to model the LR and HR image patches using Markov network [1]. Assuming a manifold structure of LR and HR image patches, locally linear embedding is applied to estimate the optimal representation using the neighbors of each LR patch [4]. For face hallucination, the idea of neighbor embedding is extended to the discrete cosine transform (DCT) space [9], and further developed to coupled-constrained neighbor embedding [10]. Different from those methods using a fixed number of neighbor for reconstruction, single-image super-resolution method based on sparse representation adaptively selects the most relevant dictionary atoms to minimize the reconstruction error [2]. Recently, the applications of deep neural networks for single-image super-resolution have produced surprisingly good effects [11], [12], [13].

Joint dictionary learning of both LR and HR dictionary is an

essential step for single-image super-resolution. Representative works include the Method of Optimal Directions (MOD) [14], the K-SVD algorithm [15] and online large-scale learning algorithms [16]. To further coupling the observed LR space and latent HR space, coupled dictionaries are learned using bilevel optimization [17]. Fisher discriminative dictionaries are learned to achieve both diversity and discriminative [18].

III. De-biased Sparse Representation

A. Background

For a given vector $\mathbf{x} \in \mathbb{R}^{m\times 1}$, it can be approximated by a sparse linear combination of atoms of an overdetermined dictionary $\mathbf{D} \in \mathbb{R}^{m\times k}, (m \ll k)$, where k is the number of dictionary atoms. Then the sparse representation is given by:

$$\alpha^{\lambda} = \arg\min_{\alpha} \frac{1}{2}\|\mathbf{x} - \mathbf{D}\alpha\|_2^2 + \lambda\|\alpha\|_1,$$

where $\|\alpha\|_1$ is the sparse inducing norm, and $\lambda > 0$ is the regularization parameter that balances data fidelity and sparsity level.

Under mild conditions of dictionary $\mathbf{D}$, ground truth α and well-chosen λ, above optimization can lead to the exact recovery of the support of ground truth α [6]. However, a series of optimization problems need to be solved with $\lambda \to 0$ to achieve an unbiased solution.

B. Proposed De-biasing Framework

The key idea of de-biasing is to add a consecutive step minimizing the data fidelity on an appropriate set:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2}\|\mathbf{x} - \mathbf{D}\alpha\|_2^2 \\ \text{s.t.} \quad & \text{Dist}(\alpha, \alpha^{\lambda}) = 0, \end{aligned}$$

where $\text{Dist}(\alpha, \alpha^{\lambda}) = 0$ defines desired feasible set of α. Using proximal gradient method, the above optimization problem can be solved.

Algorithm 1 illustrates the framework of de-biased sparse representation method. The algorithm consists of two steps: biased solution estimation and de-biasing. The de-biasing step is formed in loops by gradient descent and projection onto the feasible set, alternatively. Such proximal gradient descent method has also been successfully applied to image denosing [19], multiple task learning [20] and matrix completion [21].

C. Sign Consistency

Here, we show that sign consistency is equivalent to minimize the generalized Bregman distance $\text{Breg}^{p}_{\|\cdot\|_1}[\mathbf{u}, \mathbf{v}]$ defined on the ℓ_1 norm of $\mathbf{u}$ and $\mathbf{v}$:

$$\begin{aligned} \min_{\alpha} \quad & \frac{1}{2}\|\mathbf{x} - \mathbf{D}\alpha\|_2^2 \\ \text{s.t.} \quad & \text{Breg}^{p}_{\|\cdot\|_1}\left[\alpha, \alpha^{\lambda}\right] = 0. \end{aligned}$$

Input: Signal $\mathbf{x}$, dictionary $\mathbf{D}$ and regularization parameter λ.
Output: De-biased sparse representation α.

Estimate the biased solution:
  $\alpha^{\lambda} = \arg\min_{\alpha} \frac{1}{2}\|\mathbf{x} - \mathbf{D}\alpha\|_2^2 + \lambda\|\alpha\|_1$
Set $t = 1$.
while *convergence is not achieved* **do**
  Gradient descent:
    $\alpha^{t+1} = \alpha^{t} - \tau(\mathbf{x} - \mathbf{D}\alpha^{t})$;
  Projection α^{t+1} onto the feasible set:
    $\text{Dist}(\alpha^{t+1}, \alpha^{\lambda}) = 0$.
end

Algorithm 1: De-biased Sparse Representation using Proximal Gradient Method

Note that the generalized Bregman distance is given by

$$\text{Breg}^{p}_{J}[u, v] = J(u) - J(v) - \langle p, u - v\rangle, \quad p \in \partial J(v).$$

Thus, we have

$$\begin{aligned} \text{Breg}^{p}_{\|\cdot\|_1}\left[\alpha, \alpha^{\lambda}\right] &= \|\alpha\|_1 - \langle p, \alpha\rangle \\ &= \sum_i |\alpha_i| - p_i\alpha_i \\ &= \sum_i \left(\text{sign}(\alpha_i) - p_i\right)\alpha_i \end{aligned}$$

Note that the subdifferential of $J = \|v\|_1$ is given by

$$p_i = \begin{cases} 1 & \text{if } v_i > 0, \\ -1 & \text{if } v_i < 0, \\ [-1, 1] & \text{otherwise.} \end{cases}$$

A zero Bregman distance implies either $\alpha_i = 0$ or $\text{sign}(\alpha_i) = p_i^{\lambda}$, which further demonstrates the equivalence of feature selection consistency and generalized Bregman distance defined on the ℓ_1 norm.

The de-biasing optimization with sign consistency can be solved efficiently using the optimization method illustrated in Algorithm 1. The projection step is carried out with $\mathcal{O}(1)$ by:

$$\alpha^{k+1}[sign(\alpha^{k+1}) \odot sign(\alpha^{\lambda}) \neq 1] = 0,$$

where $\odot$ denotes the element-wise product.

D. Feature Consistency

Feature consistency is a relaxed form of sign consistency, which encourages features to have the same support, but do not necessarily to have the same sign. This consistency is equivalent to the infimal convolution between the ℓ_1 Bregman distance (ICB) [22]:

$$\begin{aligned} &\text{ICB}[\alpha, \alpha^{\lambda}] \\ =&\text{Breg}^{p}_{\|\cdot\|_1}[\alpha, \alpha^{\lambda}] \,\square\, \text{Breg}^{-p}_{\|\cdot\|_1}[\alpha, -\alpha^{\lambda}] \\ =&\sum_i (1 - |p_i|)|\alpha_i|. \end{aligned}$$

Therefore, for $\forall p_i = \text{sign}(\alpha_i^\lambda)$, a zero ICB distance implies α_i and α_i^λ have a same support.

Though feature consistency can also be solved using the unified method shown in Algorithm 1, we can replace the entire **while** loop by the following analytical solution:

$$\alpha = (\mathbf{D}_{\alpha^\lambda}^\top \mathbf{D}_{\alpha^\lambda})^{-1} \mathbf{D}_{\alpha^\lambda}^\top \mathbf{x},$$

where $\mathbf{D}_{\alpha^\lambda} \in \mathbb{R}^{m \times k'}$, and $k' \ll k$ is the number of nonzero items of α^λ.

IV. Single-Image Super-Resolution

Given sampled training image patch pairs $\{\mathbf{X}^L, \mathbf{Y}^H\}$, where $\mathbf{Y}^L = [\mathbf{y}_1 \ \mathbf{y}_2 \ \cdots \ \mathbf{y}_n]$ denotes LR patches and $\mathbf{X}^H = [\mathbf{x}_1 \ \mathbf{x}_2 \ \cdots \ \mathbf{x}_n]$ denotes the corresponding HR image patches. Our goal is to first learn paired dictionaries ($\mathbf{D}_L$ and $\mathbf{D}_H$) such that LR and HR patches can share equivalent sparse representations. After that, testing LR patches are sparse represented using LR dictionary $\mathbf{D}_L$, and testing HR patches are finally reconstructed.

The dictionary learning procedure is to optimize:

$$\begin{aligned} \min_{\mathbf{D}_c, \mathbf{A}} \quad & \|\mathbf{X}_c - \mathbf{D}_c \mathbf{A}\|_2^2 + \tau \|\mathbf{A}\|_1 \\ \text{s.t.} \quad & \|(\mathbf{D}_c)_i\|_2^2 \leq 1 \quad \forall i = 1, \cdots, K, \end{aligned}$$

where

$$\mathbf{X}_c = \begin{bmatrix} \frac{1}{\sqrt{M}} \mathbf{X}^L \\ \frac{1}{\sqrt{N}} \mathbf{Y}^H \end{bmatrix}, \mathbf{D}_c = \begin{bmatrix} \frac{1}{\sqrt{M}} \mathbf{D}_L \\ \frac{1}{\sqrt{N}} \mathbf{D}_H \end{bmatrix}.$$

.

Though above formulation is not convex in both dictionary $\mathbf{D}$ and sparse representation $\mathbf{A}$, we optimize them in an alternative manner:

Fix $\mathbf{D}_c$, optimize $\mathbf{A}$ using the de-biasing method with feature selection consistency describe in Section III-D.

Fix $\mathbf{A}$, optimize $\mathbf{D}_c$ by:

$$\begin{aligned} \mathbf{D}_c \quad & = \arg\min_{\mathbf{D}_c} \|\mathbf{X}_c - \mathbf{D}_c \mathbf{A}\|_2^2 \\ \text{s.t.} \quad & \|(\mathbf{D}_c)_i\|_2^2 \leq 1, i = 1, \cdots, K. \end{aligned}$$

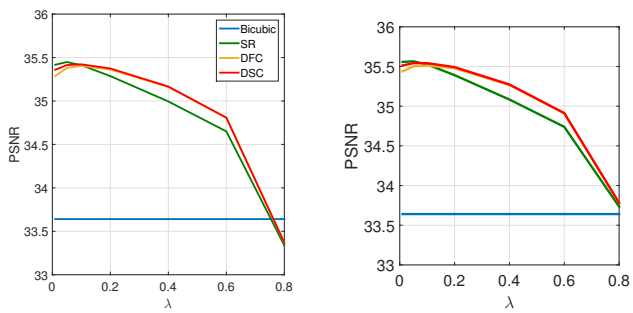
In the testing stage, for test LR image patch set $\mathbf{X}_{test}$, we apply above mentioned de-biased methods to evaluate sparse representation $\mathbf{A}$.

Finally, the super-resolved image patches $\hat{\mathbf{Y}}$ are constructed using HR dictionary $\mathbf{D}_H$ and sparse representation $\mathbf{A}$:

$$\hat{\mathbf{Y}} = \mathbf{D}_H \mathbf{A}.$$

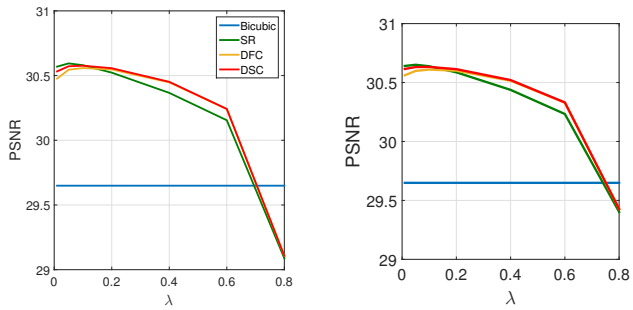
V. Experimental Results

In this section, we describe the details of experiments to evaluate the effectiveness of the de-biasing methods for single-image super-resolution.



(a) (b)

Fig. 1. 2× SR for averaged PSNR evaluation of Bicubic, sparse representation (SR), de-biasing with feature consistency (DFC) and de-biasing with sign consistency (DSC) on **Set5** by varying the regularization parameter λ with dictionary size: (a) 512; (b) 1024.



(a) (b)

Fig. 2. 2× SR for averaged PSNR evaluation of Bicubic, sparse representation (SR), de-biasing with feature consistency (DFC) and de-biasing with sign consistency (DSC) on **Set14** by varying the regularization parameter λ with dictionary size: (a) 512; (b) 1024.

A. Datasets and Settings

We compare the performance on two commonly used datasets, i.e., **Set5** [23] and **Set14** [24]. Set5 contains five images: Baby, Bird, Butterfly, Head and Woman. Set14 contains fourteen images: Baboon, Barbara, Bridge, Coastguard, Comic, Face, Flowers, Foreman, Lenna, Man, Monarch, Pepper, PPT3 and Zebra.

For each color image, we first convert it to YCbCr space. Only Y channel is super-resolved, and Cb, Cr channels are interpolated using bicubic interpolation. The patch size is 5×5 and the overlap between neighbor patches is 4 pixels. The final results of overlapped regions are evaluated by averaging.

Peak signal-to-noise ratio (PSNR) and Structural Similarity Index (SSIM) are measure as performance measures. To be fair to all methods, we do not perform back projection as a post-processing for all the experiments.

B. Performance

We first compare the PSNR evaluation on Set5 and Set14 for 2× SR. Different dictionary with varying regularization parameter λ in (0.001 0.05 0.1 0.2 0.4 0.6 0.8) are tested. The PSNR curves for Set5 and Set14 are shown in Figure

Table 1. $2\times$ SR for PSNR and SSIM evaluations of Bicubic, sparse representation (SR), de-biasing with feature consistency (DFC) and de-biasing with sign consistency (DSC) using various λ.

Measure	λ	Bicubic	SR	DFC	DSC
PSNR	0.05	32.79	34.57	34.48	34.52
	0.2	32.79	34.50	34.52	34.54
	0.4	32.79	34.27	34.39	34.40
SSIR	0.05	0.9614	0.9694	0.9688	0.9691
	0.2	0.9614	0.9698	0.9689	0.9691
	0.4	0.9614	0.9676	0.9683	0.9683

1 and Figure 2, respectively. We notice the gap between the proposed two de-biased methods (DFC and DSC) and the original sparse representation method (SR) are significant and consistent, especially λ is in the range of 0.2-0.6. We also note that the PSNR evaluation of the original method is better than our methods for λ ranging from 0.01 to 0.05. This phenomenon is due to the rising variance for our methods with extremely small regularization parameters.

To give a more detailed comparison, we use the Lenna image for more visual and quantitative analysis. According to the curves in Figure 1 and Figure 2, we compare the results with $\lambda = 0.05, 0.2$ and 0.4. Figure 3 illustrates the $2\times$ super-resolution visual results for sparse representation (SR), de-biasing with feature consistency (SFC), de-biasing with sign consistency (DSC) using various λ. Compared to SR, the proposed methods can faithfully reconstruct the edges and detailed texture features. The corresponding PSNR and SSIM evaluations in Table 1 also corroborate the superiority of the proposed methods, especial when λ is large.

VI. Conclusion

We have presented a de-biasing framework to improve the performance of single-image super-resolution based on sparse representation. Under this framework, two de-biasing methods with sign consistency and feature consistency are proposed. Experimental results on two real image datasets verified the effectiveness and robustness of our methods. One future work is to balance the bias and variance instead of merely de-biasing for extremely small regularization parameters.

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(a) LR (b) Bicubic (c) HR

(d) $\lambda = 0.05$

(e) $\lambda = 0.2$

(f) $\lambda = 0.4$

Fig. 3. $2\times$ SR for visual comparison between sparse representation, de-biasing with feature consistency, de-biasing with sign consistency from left to right for subfigure (d)-(f).